MIL-G-60143A(W)
11 July 1966
SUPERSEDING
MIL-G-(60143(MU)
11 January 1965

MILITARY SPECIFICATION

GYROSCOPES, RATE: 10527237-1 and 10527237-2

1. SCOPE

1.1 This specification covers two types of rate gyroscopes that function in a gyro-stabilized sight assembly employed in a UH-lB helicopter. These gyroscopes are essentially identical except for damping factor (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	*Sampling Procedures and Tables for Inspection by Attributes*
MIL-STD-109	*Quality Assurance Terms and Definitions*

DRAWINGS

U.S. Army, Frankford Arsenal

D10527237	*Gyroscopes, Rate: 10527237-1, 10527237-2*

PACKAGING DATA SHEET

MIL-P-14232/P10527237 Packaging of Gyroscope, Rate: 10527237

FSC 1270

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submittal of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The rate gyroscopes shall be manufactured in accordance with Drawing D10527237 and drawings pertaining thereto.

3.2.1 Treatment, moisture and fugus -resistant.- The rate gyroscopes shall be treated for moisture and fungus resistance In accordance with MIL-T-152.

3.2.2 Insulation resistance.- The insulation resistance between isolated electrical circuits, and between the gyroscope case and circuits isolated from the case shall be not less than 10 megohms when 100 ± 10 volts direct current (Vdc) is applied for 10 ± 2 seconds (sec).

3.2.3 Dielectric strength.- The leakage current shall not exceed 1.0 milliampere (ma) when 115 ± 10 volts root mean square (Vrms), at 400 cycles per second (cps) is applied between isolated electrical circuits, and between the gyroscope case and circuits isolated from the case for a period of 10 ± 2 sec.

3.3 General speciFication.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part, identification and marking
(d) Workmanship
(e) Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temperature.- The rate gyroscopes shall meet the performance requirements of 3.5 at 60° to 90° Fahrenheit (F) after being subjected to ambient temperatures at -80° and at +160°F.

3.4.2 Operating temPerature.- The rate gyroscopes shall meet the performance requirements of 3.5 while exposed to-temperatures at -40° and at +150° Fahrenheit (F).

3.4.3 Altitude.- The rate gyroscopes shall meet the performance requirements of 3.5 at an altitude of 30,000 feet above sea level.

3.4.4 Shock.- The rate gyroscopes shall withstand impact shocks up to 15 G's, each impulse having a time duration of 11 ± 1 milliseconds. Subsequently, the rate gyroscopes shall meet the performance requirements of 3.5.

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3.4.5 Vibration.- The rate gyroscopes shall meet the performance requirements of 3.5 after being subjected to vibrations over a frequency range of 5 to 500 cycles per second (cps). For frequencies between 5 to 20 cps, the double amplitude shall be 1.0" D.A. For frequencies between 20 to 33 Cps, the amplitude shall vary from 1 inch (l") D.A. at 20 cps to .67" at 33 cps. For frequencies between 33 to 50 cps, the amplitude shall be varied so as to maintain a 35G acceleration. For frequencies between 50 to 95 Cps, the amplitude shall be varied so as to maintain a 50G accelel tion. For frequencies between 95 to 500 CPS, the amplitude shall be varied so as to maintain a 10G acceleration.

3.5 Performance. (Types 1 and 2 except where otherwise indicated)

3.5.1 Rate transfer characteristics.

3.5.1.1 Rate range.- Rate range shall be at least ± 40° per sec.

3.5.1.2 Sensitivity.- Nominal sensitivity shall be 140 millivolts root mean square per degree per second (mv rms/deg/see) ± 10% over the temperature range (-40° to 150°F) when excited by a 26-volt (v) 400 cps source in series with a 50 millihenry (MH) choke having a Q of 15 and the pick-off loaded with a 10K resistor.

3.5.1.3 Linearity.- Linearity of the rate gyroscopes shall be ± 0.5% of full rate scale to half rate scale. It shall be ± 3.0% of full rate scale across the full rate scale output. Linearity shall be the maximum deviation of the calibration curve from the most favorable straight line that can be drawn through the calibration curve and which coincides with the calibration curve for zero rate input.

3.5.1.4 Threshold.- Threshold shall be less than 0.005 deg/sec.

3.5.1.5 Hysteresis.: Hysteresis shall be less than ± 0.06 deg/sec.

3.5.2 Dynamic response characteristics.

3.5.2.1 Damping factor (Part NO. 10527237-l) .- The nominal damping factor, expressed as a ratio at any given temperature to critical damping of the gyroscope natural frequency, shall be 0.7 ± 30% over the environmental temperature range.

3.5.2.2 Damping factor (Part No. 10527237-2) .- The nominal damping factor, expressed as a ratio at any given temperature to critical damping of the gyroscope natural frequency shall be 5.0 ± 2,0.

3.5.2.3 Natural frequencY.- Natural frequency shall be at least 25 Cps.

3.5.3 Pickoff phase shift.- The phase shift of the output signal with

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respect to the 26V, 400 cps excitation shall be 7 ± 5° with the pickoff loaded by a 10K load.

3.5.4 *Run-up time.* - *The rate" gyroscope shall come up to synchronous speed in less than 28 seconds over the environmental temperature range.*

3.5.5 *Modulation noise.*- *The modulation or residual noise shall not be greater than .06°/sec peak-to-peak over the frequency range from 5 to 100 Cps. For the frequencies between O and 5 cps, the modulation noise shall not be greater than .03°/sec peak-to-peak.*

3.5.6 *Zero set.*- *The* zero set of the rate gyroscopes shall be less than .06°/sec.

3.6 Radio interference suppression.- Interference control requirements of the equipment shall conform to MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable. .

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

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4.2.3 *First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.*

4.3 Inspection provisions.

4.3.1 Submission of Product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-sTD-105 .

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

TABLE I

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Altitude	3.4.3	4.6.2.3
302.	Shock	3.4.4	4.6.2.4
303.	Vibration	3.4.5	4.6.2.5
304.	Radio interference suppression	3.6	4.6.4

4.3,2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables 11 and *111* herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables 11 and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE 11 - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR : AQL Percent Defective		
101. *Treatment, moisture and fungus resistant*	3.2.1	*4.6.1*

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TABLE II - CLASSIFICATION OF DEFECTS (Cent'd)

		REQUIREMENT	TEST PROCEDURE
102.	Insulation resistance	3.2.2	4.6.1.1
103.	Dielectric strength	3.2.3	4.6.1.2
104.	Rate transfer characteristics	3.5.1	4.6.3.1
105.	Dynamic response characteristics	3.5.2	4.6.3.2
106.	Pickoff phase shift	3.5.3	4.6.3.3
107.	Modulation noise	3.5.5	4.6.3.5
108.	Zero set	3.5.6	4.6.3.6

MINOR: NONE

4.3.3 *Acceptance and rejection.- Rejected lots. shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.*

4.4 Special sampling.- Three gyroscopes shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE 111

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Fabrication	3.2	Applicable drawing - visual
402.	General specification	3.3	MIL-F-13926 - visual
403.	Temperature (storage)	3.4.1	4.6.2.1
404.	Temperature (operating)	3.4.2	4.6.2.2

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with "the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

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4.5.2.1 *Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).*

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Treatment, moisture and fungus resistant.- The rate gyroscope shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.1.1 Insulation resistance test.- Using a hi-pot, apply 100 ± 10 Vdc between isolated electrical circuits, and between the gyroscope case and circuits isolated from the case. Check each circuit for a period of 10~ 2 sec. Calculate the insulation resistance to insure compliance with 3.2.2.

4.6.1.2 Dielectric strength test.- Using a hi-pot (a-c type), apply ll5 ± *10 v* rms, 400 cps between isolated electrical circuits, and between the gyroscope case and circuits isolated from the case. Check each circuit for a period of 10 ± 2 sec. Verify leakage current to insure compliance with *3.2.3.*

4.6.2 Environmental.

4.6.2.1 Storage temperature.- The gyro, while in a non-operating condition, shall be subjected to the Temperature Cycling Test, *Procedure 11 of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the gyro shall meet the performance requirements specified in 3.5.*

4.6.2.2 Operating temperature.- The gyro shall be subjected to-the Temperature Cycling Test, Procedure I of MIL-F-13926 using Conditions C and G. During temperature stabilization at Conditions C, G and room ambient, the gyro shall meet the performance requirements specified in 3.5.

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line shall comply with 3.5.1.2. The linearity of the test data with respect to the best straight line shall comply with "3.5.1.3. *The test point at 40°/sec meeting the requirements of 3.5.1.3 shall verify rate range per 3.5.1.1. The threshold, the total value of rate input on the abscissa for zero pickoff output, shall comply with 3.5.1.4.*

4.6.3.2 Dynamic response characteristics test.- The *following test equipment is required for this test:*

(a) Single axis oscillation platform, base and mounting plate.
(b) Means of applving sinusoidal oscillating inputs to the gyroscope at-known variable frequency and amplitude.
(c) Means of reading the phase angle between gyroscope input (table rate) and output (pins 2 and 3).

Energize the gyroscope according to 4.6.3.1. Connect the gyroscope pickoff and table angular position pickoff to a phase angle measuring device. Measure the frequency at which the phase angle of the gyroscope output lags the input by 90° (natural frequency). The natural frequency shall be at least 25 cps per 3.5.2.3. Measure the phase angle of the 10527237-1 gyro at one-half the 90° frequency previously determined. The measured phase angle at this half frequency shall be 42° ± 10° to insure compliance with 3.5.2.1. Measure the frequency of the 10527237-2 gyro at which a 45° phase lag occurs. The frequency will be 1.9 to 4.4 cps. This is equivalent to a damping ratio of 5.0 ± 2.0 with a natural frequency of 26.5 CpS. To comply with 3.5.2.2.

4.6.3.3 Pickoff phase shift test.- The following test equipment is required for this test:

(a) Single axis rate table
(b) Means of measuring output phase shift with respect to excitation.
(Phase meter)

Mount the gyroscope on the rate table with its input axis vertical. *Energize the gyro according to 4.6.3.1. Measure the phase difference at cw and ccw rates of 20 deg/sec for compliance with 3.5.3.*

4.6.3.4 Modulation noise.- The total gyro output signal is phase detected and passed through a 0 to 5 cps low pass filter. The output read on a standard VTVM shall be in conformance with 3.5.5. The total gyro output signal is phase detected and passed through a 5 to 100 cps band pass filter. The output read on a standard VTVM shall be in conformance with 3.5.5.

4.6.3.5 Zero set.- Energize motor and allow to reach synchronous speed. Upon 'removal of motor power and while motor is coasting to a stop, manually apply alternatively clockwise and counterclockwise rates to the turntable. The applied rates should be less than nominal full scale, and should be of approximately constant rate amplitudes (this procedure is used to eliminate hysteresis). Gyro is then placed on end (this eliminates any gimbal unbalance), gimbal axis vertical and the motor

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energized. , The D.C. output of the gyro shall be adjusted (fine null adjustment) to a value within specification limits. The zero set shall conform to 3.5.6.

4.6.4 *Radio interference and suPPression.- The equipment shall be tested in accordance with MIL-I-6181 to insure compliance with 3.6.*

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527237 . The level of protection shall be as specified in the procurement document.

6. *NOTES*

6.1 Intended use.- The Rate Gyroscopes: 10527237-1 and 10527237-2 are part of the Antioscillation Sighting System: XM58 employed in a UH-lB helicopter. The rate gyroscopes have two-phase hysteresis synchronous motors operating from a single phase 400 cps supply with one microfarad capacitors in series with the quadrature phase inputs. The rate gyroscopes,-when mounted on the elevation gimbal, are oriented to sense *all angular motions except those about the line of sight.*

6.2 Ordering data.- Procurement documents should specify the following:

(a) *Title, number and date of this specification.*
(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

Custodian:

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Preparing activity:

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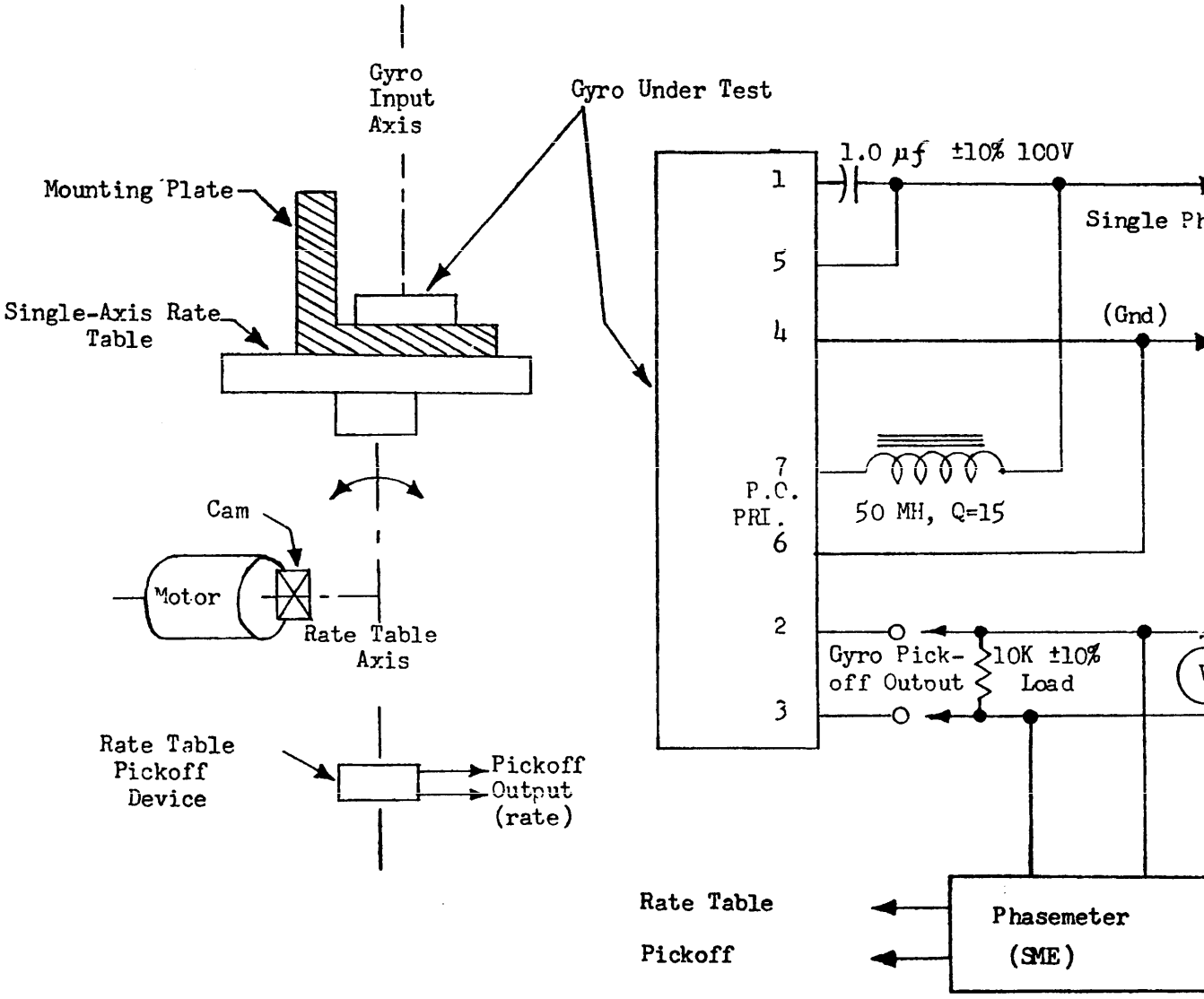


Figure 1. Test Set-up for Rate Gyroscope: 10527237

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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